

WASP-12b

R-0324

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231229 · created 2026-07-18T03:04:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460307.6861 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.03 +/- 0.019
Transit depth (Rp/Rs)^2	0.092 +/- 0.12 [%]
Orbital Inclination (inc)	81.35 +/- 2.88
Airmass coefficient 1 (a1)	1.011 +/- 0.012
Airmass coefficient 2 (a2)	-0.0102 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	#3 - [435, 328]
Optimal Aperture	3.76
Optimal Annulus	13.0
Transit Duration (day)	0.1064 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.03 ± 0.019 vs 0.1178 (deviation 4.32σ)
Duration fit vs published	0.1064 d vs 0.125 d (deviation 1.83σ)
Mid-time O–C	-3.1 min
Depth SNR	1.5
Residual scatter	1.18 %

Light curve

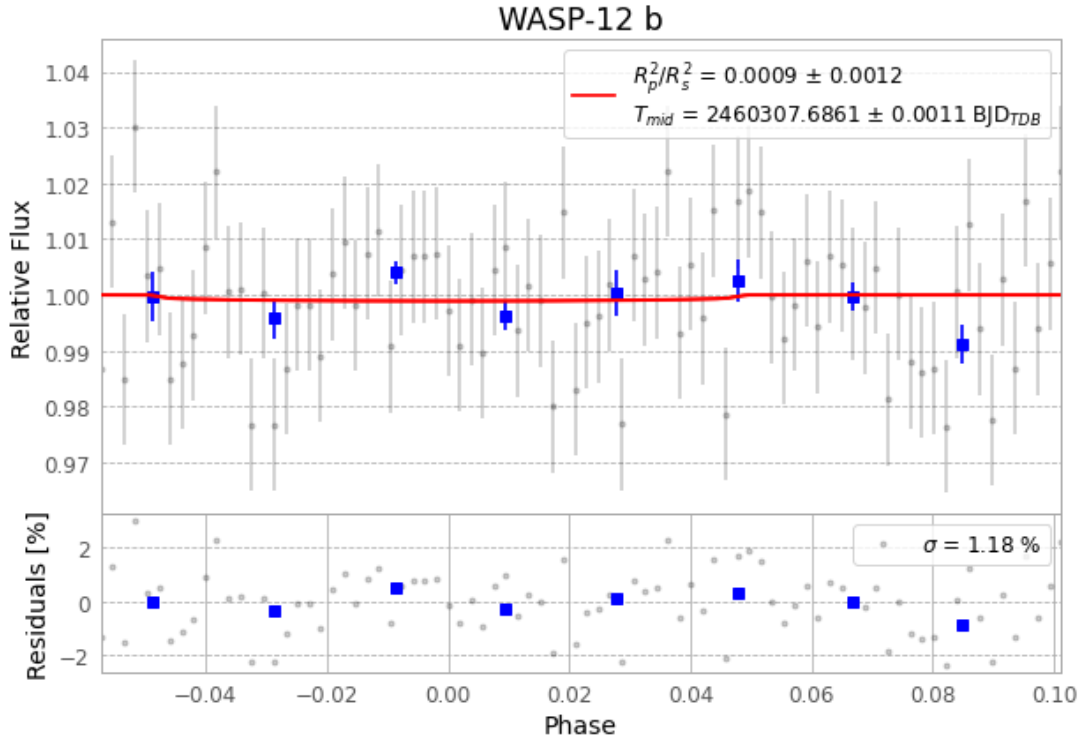


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-28.png (SHA-256 874d7202edf6a203...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a9e9321ad110cdef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

84 FITS frames (55 MB), WASP-12231229024815.FITS → WASP-12231229065715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231229.log (9ae53f5fd86d...), FinalLightCurve_WASP-12 b_2023-12-28.png (874d7202edf6...), FinalLightCurve_WASP-12 b_2023-12-28.csv (2a9c9789fc5a...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 03:04Z.